(1) (i) $\vec{\nabla}\cdot(\vec{U}\times\vec{V}) = \vec{V}\cdot(\vec{\nabla}\times\vec{U}) - \vec{U}\cdot(\vec{\nabla}\times\vec{V})$ (1)

LHS:

$$\vec{V}\cdot(\vec{\nabla}\times\vec{U}) - \vec{U}\cdot(\vec{\nabla}\times\vec{V})$$

$$= \vec{V}\cdot\left\{\left(\frac{\partial U_z}{\partial y} - \frac{\partial U_y}{\partial z}\right)\hat{i} + \hat{j}\left(\frac{\partial U_x}{\partial z} - \frac{\partial U_z}{\partial x}\right) + \hat{k}\left(\frac{\partial U_y}{\partial x} - \frac{\partial U_x}{\partial y}\right)\right\}$$

$$- \vec{U}\cdot\left\{\left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z}\right)\hat{i} + \hat{j}\left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x}\right) + \hat{k}\left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y}\right)\right\}$$

$$= -V_y\frac{\partial U_z}{\partial x} + V_z\frac{\partial U_y}{\partial x} + \cdots \text{ other contributions from } \hat{j}\cdot\hat{j} \text{ \& } \hat{k}\cdot\hat{k}$$

$$- \left\{-U_y\frac{\partial V_z}{\partial x} + U_z\frac{\partial V_y}{\partial x} + \cdots \text{ ditto.} \right\}$$

$$= \frac{\partial}{\partial x}(U_y V_z - V_y U_z) + \cdots$$

$$= \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)\left\{(U_y V_z - V_y U_z)\,\hat{i} + \cdots\right\}$$

$$= \vec{\nabla}\cdot(\vec{U}\times\vec{V})$$

LHS = RHS

(ii) $\left(\vec{\nabla} \times (\nabla T)\right)$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial T}{\partial x} & \frac{\partial T}{\partial y} & \frac{\partial T}{\partial z} \end{vmatrix}$$

$$= \hat{i}\left(\frac{\partial^2 T}{\partial y \partial z} - \frac{\partial^2 T}{\partial z \partial y}\right) - \hat{j}\left(\frac{\partial \partial T}{\partial x \partial z} - \frac{\partial \partial T}{\partial z \partial x}\right) + \hat{k}\left(\frac{\partial^2 T}{\partial x \partial y} - \frac{\partial^2 T}{\partial y \partial x}\right)$$

$$= \hat{i} \times 0 - \hat{j} \times 0 + \hat{k} \times 0 = 0$$

$f(x, y, z) = x^2 y^3 z^4$

$\frac{\partial f}{\partial x} = 2x y^3 z^4$, $\frac{\partial f}{\partial y} = 3x^2 y^2 z^4$, $\frac{\partial f}{\partial z} = 4x^2 y^3 z^3$

$\frac{\partial^2 f}{\partial y \partial x} = 6x y^2 z^4$, $\frac{\partial^2 f}{\partial x \partial y} = 6x y^2 z^4$, $\frac{\partial^2 f}{\partial x \partial z} = 8x y^3 z^3$

$\frac{\partial^2 f}{\partial z \partial x} = 8x y^3 z^3$, $\frac{\partial^2 f}{\partial z \partial y} = 12 x^2 y^2 z^3$, $\frac{\partial^2 f}{\partial y \partial z} = 12 x^2 y^2 z^3$

by comparing $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$

$\frac{\partial^2 f}{\partial x \partial z} = \frac{\partial^2 f}{\partial z \partial x}$ $\Rightarrow \vec{\nabla} \times (\nabla f) = 0$

$\frac{\partial^2 f}{\partial y \partial z} = \frac{\partial^2 f}{\partial z \partial y}$

1 (iii) $\nabla^2 T = \nabla^2 e^{-5x} \sin 4y \cos 3z$

$$= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) T$$

$$= 25 e^{-5x} \sin 4y \cos 3z - 16 e^{-5x} \sin 4y \cos 3z - 9 e^{-5x} \sin 4x \cos 3z$$

$$= 0$$

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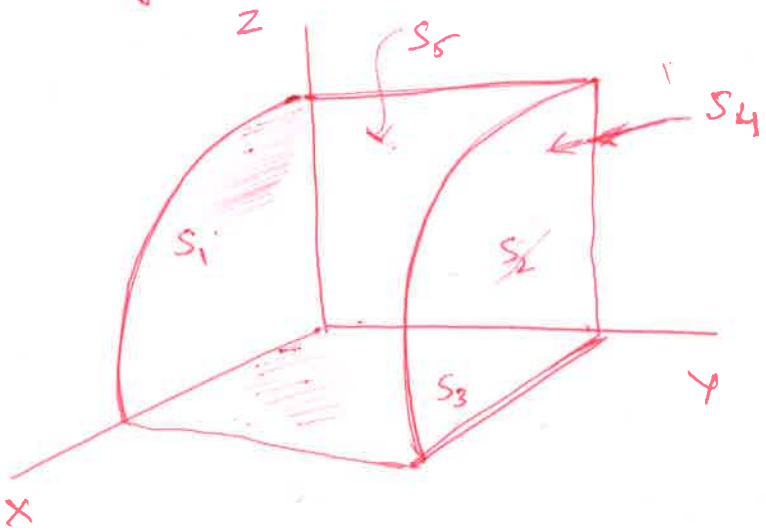
2 $\vec{F} = 6z\hat{i} + (2x+y)\hat{j} - x\hat{k}$ | $x^2+z^2 \leq 9$, $x \geq 0$, $z \geq 0$, $0 \leq y \leq 8$

$S_1 \equiv y = 0$ $\quad S_3 \equiv z = 0$

$S_2 = y = 8$ $\quad S_4 \equiv x = 0$ $\quad S_5 \equiv x^2+z^2 = 9$

$\vec{\nabla} \cdot \vec{F} = 1$



$$\int \vec{\nabla} \cdot \vec{F}\, d\tau = \int 1\, dx\, dy\, dz$$

$$= \frac{1}{4} \pi r^2 h$$

$$= \frac{1}{4} \pi\, 9 \times 8$$

$$= 18\pi$$

$$\oint_S \vec{F} \cdot d\vec{s} = \int_{S_1} \vec{F} \cdot d\vec{s} + \int_{S_2} \vec{F} \cdot d\vec{s} + \int_{S_3} \vec{F} \cdot d\vec{s} + \int_{S_4} \vec{F} \cdot ds + \int_{S_5} \vec{F} \cdot ds$$

$$= \int_{S_1} \vec{F} \cdot dx\,dz(-\hat{j}) + \int_{S_2} \vec{F} \cdot dx\,dz\,\hat{j} + \int_{S_3} \vec{F} \cdot dx\,dy(-\hat{k}) + \int_{S_4} \vec{F} \cdot dy\,dz(-\hat{i})$$

$$+ \int_{S_5} \vec{F} \cdot \frac{(2x\hat{i} + 2z\hat{k})}{2z}\, dx\,dy$$

$$= \int_{S_1} (2x+y)\,dx\,dz + \int_{S_2} (2x+y)\,dx\,dz + \int_{S_3} x\,dx\,dy +$$

$$- \int 6z\,dy\,dz + \int \left(\frac{12xz + 2zx}{2z}\right) dx\,dy$$

$$= -\int_{S_1} 2x\,dx\,dz + \int_{S_2} (2x+8)\,dx\,dz + \int_{S_3} x\,dx\,dy$$

$$-\int_{S_4} 6z\,dy\,dz + \int 5x\,dx\,dy$$

$$= -\cancel{\int_{S_1} 2x\,dx\,dz} + \cancel{\int_{S_2} 2x\,dx\,dz} + \int_{S_2} 8\,dx\,dz + \cancel{\int_0^8\int_0^3 x\,dx\,dy}$$

$$= 6\cancel{\int_0^8\int_0^3 z\,dz\,dy} + 5\int_0^8\int_0^3 \cancel{x}\,dx\,dy$$

$$= 8\int dx\,dz = 8 \times \frac{1}{4}\pi \times 9$$

$$= 18\pi$$

3 $\vec{V} = (x^2y - xy^2)\hat{i} + \left(\frac{y^3}{3} - xy^2\right)\hat{j}$

$$\vec{\nabla}\times\vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x^2y - xy^2) & \frac{y^3}{3} - xy^2 & 0 \end{vmatrix}$$

$$= \hat{k}\left(-y^2 - x^2 + 2xy\right) = -(x-y)^2\,\hat{k}$$

Statement $\int \vec{\nabla}\times\vec{V}\cdot d\vec{s} = \oint_C \vec{V}\cdot d\vec{l}$

$$\int \vec{\nabla}\times\vec{V}\cdot d\vec{s} = -\int (x-y)^2\,dx\,dy = \int -x^2\,dx\,dy - \int y^2\,dx\,dy + \int 2xy\,dy\,dx$$

$$= -\int x^2\,dx\,dy = -\int_0^1\int_0^x x^2\,dy\,dx - \int_0^1\int_0^x y^2\,dy\,dx + \int_0^1\int_0^x 2xy\,dy\,dx$$

$$= -\int_0^1 x^2 y\Big|_0^x dx - \int_0^1 \frac{y^3}{3}\Big|_0^x dx + \int_0^1 2x\frac{y^2}{2}\Big|_0^x dx$$

$$= -\cancel{\int_0^1 x^3 dx} - \int_0^1 \frac{x^3}{3}dx + \cancel{\int_0^1 x^3 dx}$$

$$= -\frac{1}{3}\times\frac{x^4}{4}\Big|_0^1 = -\frac{1}{12}$$

$$\int \vec{V}\cdot d\vec{l} = \int_{C_1} \vec{V}\cdot dx\,\hat{i} + \int_{C_2} \vec{V}\cdot dy\,\hat{j} + \int_{C_3} \vec{V}\cdot(\hat{i}+\hat{j})\,dy$$

$$= \int_0^1 (x^2y - y^2x)\,dx + \int_0^1 \left(\frac{y^3}{3} - xy^2\right)dy + \int_1^0 (x^2y - xy^2)\,dy + \int_1^0 \left(\frac{y^3}{3} - xy^2\right)dy$$

$$= 0 + \int_0^1 \left(\frac{y^3}{3} - y^2\right) dy + \int_1^0 \left(\cancel{y^3 - y^3}\right) dy$$

$$C_2$$

$$+ \int_1^0 \left(\frac{y^3}{3} - yy^2\right) dy$$

$$= -\int_0^1 y^2 dy - \int_1^0 y^3 dy$$

$$= -\left|\frac{y^3}{3}\right|_0^1 - \left|\frac{y^4}{4}\right|_1^0$$

$$= -\frac{1}{3} + \frac{1}{4} = \frac{-4+3}{12} = -\frac{1}{12}$$